

Scalability & Future Plan

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Voice-Based Concept Understanding Analyser can be scaled to support a larger number of users, improved AI models, enhanced analytics, and cloud-based deployment. It also presents the future roadmap for improving the application's performance, security, and user experience.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports limited concurrent users	Performance may decrease during heavy usage	High
2	Audio files are processed sequentially	Increased response time for multiple requests	Medium
3	Limited concept knowledge base	Evaluation accuracy depends on available reference concepts	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports small number of users	Deploy on cloud with auto-scaling and load balancing
2	Data Storage	Local storage	Migrate to cloud databases and object storage
3	Performance	Single-server processing	Optimize AI pipeline, caching, and parallel audio processing
4	Security	Basic authentication	Implement OAuth, encrypted storage, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Multi-language speech recognition	2 Months	Support users speaking different languages
Phase 2	Personalized AI feedback and learning recommendations	4 Months	Improve student learning experience
Phase 3	Teacher/Admin Dashboard with analytics and progress tracking	6 Months	Better monitoring of student performance
Phase 4	Cloud deployment, mobile application, and LMS integration	8 Months	High scalability, accessibility, and wider adoption